

Work Breakdown Structure — PD-RPM Remote Care System

This work breakdown structure decomposes the PD-RPM remote care system into eight level-two elements and twenty-eight work packages. Elements 1.1–1.4 deliver the product itself, following the path of data from the home to the clinical team; elements 1.5–1.8 produce no hardware but consume schedule, budget, and staff, and are carried so the plan resources them.

1.1 Home Data Acquisition

- 1.1.1 Cyclor Data Interface Module
- 1.1.2 Vitals Peripherals (BP, scale)
- 1.1.3 Patient Symptom & Exit-Site App
- 1.1.4 Home Gateway & Store-Forward

1.2 Data Transport & Integration

- 1.2.1 Secure Transport & Encryption
- 1.2.2 EHR Integration (HL7/FHIR)
- 1.2.3 Ingestion & Data Normalization

1.3 Clinical Monitoring

- 1.3.1 Trending & Analytics Engine
- 1.3.2 Alert & Threshold Logic
- 1.3.3 Clinician Dashboard & Triage Queue
- 1.3.4 Escalation & Notification Service

1.4 Sustainment & Support

- 1.4.1 Device Logistics & Provisioning
- 1.4.2 Patient / Care Partner Training
- 1.4.3 Field & Technical Support Tooling

1.5 Systems Engineering & Management

- 1.5.1 Requirements & Architecture
- 1.5.2 Interface Management
- 1.5.3 Trade Studies & Design Analysis
- 1.5.4 Configuration & Data Management

1.6 Integration, Test & Validation

- 1.6.1 Component & Subsystem Test
- 1.6.2 System Integration
- 1.6.3 Clinical Usability Validation
- 1.6.4 Cybersecurity & Penetration Test

1.7 Risk Management

- 1.7.1 Risk Identification
- 1.7.2 Analysis & Mitigation Planning
- 1.7.3 Risk Monitoring & Reporting

1.8 Regulatory & Quality Assurance

- 1.8.1 SaMD Classification & FDA Path
- 1.8.2 HIPAA Privacy & Security
- 1.8.3 QMS & Design History File

Outside the WBS

Required for operation but not built by this project; an interface to each is defined instead.

- **PD cyclor** — manufacturer’s regulated Class II device; interface defined at 1.1.1
- **Broadband / cellular carrier network** — interface defined at 1.2.1
- **EHR platform itself** — interface defined at 1.2.2
- **Dialysate manufacturing & distribution** — interface defined at 1.4.1